JACOBSONNESS READ OFF THE CLOSED POINTS OF A KOLMOGOROV SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The equivalence between Jacobsonness and a condition on locally closed subsets through closed points. Invoked at the start of the counted proof (source line 3733). Reduces the assertion to a retrocompactness comparison.
- The statement that the closed points are inherited by open subsets. Invoked inside the counted proof (source lines 3737-3738). Allows replacing the space by a given open subset.

Source reference. Original lemma lemma-jacobson-equivalent-locally-closed: sources/stacks-topology/pinned-original-source.tex, source lines 3694–3694. Navigation pointer only: the counted text refers to the Stacks item lemma-jacobson-equivalent-locally-closed, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-jacobson-inherited: sources/stacks-topology/pinned-original-source.tex, source lines 3642–3642. Navigation pointer only: the counted text refers to the Stacks item lemma-jacobson-inherited, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Suppose X is a Jacobson topological space. Let X_0 be the set of closed points of X. There is a bijective, inclusion preserving correspondence*

$$\{\textit{constructible subsets of } X\} \leftrightarrow \{\textit{constructible subsets of } X_0\}$$

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given by $E \mapsto E \cap X_0$. This correspondence preserves the subset of retrocompact open subsets, as well as complements of these.

Proof. From Lemma lemma-jacobson-equivalent-locally-closed above, we just have to see that if U is open in X then $U \cap X_0$ is retrocompact in X_0 if and only if U is retrocompact in X. This follows if we prove that for U open in X then $U \cap X_0$ is quasi-compact if and only if U is quasi-compact. From Lemma lemma-jacobson-inherited it follows that we may replace X by U and assume that $U = X$. Finally notice that any collection of opens $\mathcal{U}$ of X cover X if and only if they cover X_0, using the Jacobson property of X in the closed $X \setminus \bigcup \mathcal{U}$ to find a point in X_0 if it were nonempty. $\square$